

Guillaume Pelletier
DotVision, 75008 Paris, France
guillaume.pelletier@dotvision.com

Subject: Submission of manuscript to Sensors

Dear Editor,

Please find enclosed our manuscript "*Envelope-Domain Preprocessing for Ultra-Compact LSTM-Based Broken Rotor Bar Severity Classification*" for consideration in *Sensors*.

Detecting broken rotor bars in induction motors typically requires either large convolutional networks operating on spectrograms or explicit frequency-domain feature engineering with prior knowledge of slip and operating conditions. Both approaches are difficult to deploy on resource-constrained sensing nodes. This paper takes a different route: a half-cycle moving RMS extracts the current envelope, per-window centering removes the load baseline, and a 4,773-parameter LSTM classifies the result into five severity levels.

On the public Universidade Federal de Uberlândia dataset, this reaches 78.3% on 5-class classification and 94.0% on binary fault detection. The full model fits in 19 KB. Training runs in a web browser in under three minutes, with no server or GPU.

The paper includes an ablation across four pipeline variants (16% to 78%) showing that each preprocessing step contributes measurably, and a classical FFT + SVM baseline at 81.5%. The SVM scores higher but requires hand-crafted spectral features; the LSTM works directly from the time-domain envelope.

The target deployment is an ESP32-class sensing node that monitors harmonic signatures for change, then loads specialized ONNX classifiers at runtime to identify specific faults. The work sits at the intersection of signal processing for condition monitoring and embedded AI for industrial sensing, which we believe fits the scope of *Sensors*.

This manuscript is original and is not under consideration elsewhere.

Sincerely,
Guillaume Pelletier